

Housing Prices in California in 1990

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Abstract:

This paper uses California housing data to create a regression that can predict the mean of the median house value in 1990 of California houses in the neighborhoods examined. We do so a simple regression model with multiple predictor variables. Of more value, the regression also examines the impact that those predictors individually have on housing prices. It can be used to examine the impact that ocean proximity, housing median age, median income, total number of rooms, total number of bedrooms had on the pricing of houses in that region in 1990. Our findings conclude factors like the house being inland and an increase in the total number of rooms had a negative mean effect on the mean median house value, while predictors like being near the ocean and a higher median income of neighbors had a positive effect on the mean median house value.

Introduction:

This paper builds a linear regression model of the mean of the median house values using a dataset composed of approximately 20,000 houses. When considering buying a house, usually there are many factors that determine the price of a house which do not weigh equally in determining the house value. This paper presents a modeling process for estimating the median house values using a linear regression model based on several predictor variables given in our dataset. We are investigating how factors such as proximity to the ocean or the number of bedrooms in a 'block' alters housing value, holding all else constant. While this dataset is only applicable to houses from the particular neighborhood in California chosen in 1990 and thus cannot predict housing prices today, it can help us better understand the housing market in those particular neighborhoods in 1990. We put a greater focus then on quantifying the effects of these predictors and examining how well they correlate to housing prices individually as compared to together (in our regression model).

Data:

The data in this study has been collected from the 1990 California Census data and put together by Luis Torgo from the University of Porto. The data contains one row per census block group which is the geographical unit used in the Census. A block typically contains between 600 to 3000 people (the mean number of people in a block in this sample is 1425.5 people). There were some modifications made to this dataset before we began to work with it. The first is that an ocean proximity variable was added which identified whether the specified block was close to the ocean, the bay, inland, or on an island. Additionally, there were 207 values randomly removed from the `total_rooms` column. We were not able to find any explanation of why that happened. When looking at the original data it appears like there are a lot of median household values that were marked as \$500,001. We do not know why this may be but we suspect that it has to do with how census data was reported in 1990 (where the average price of a house was \$73,300, so a price above \$500,000 was extremely high) (Demographia). Finally, all rows missing data values were removed. This resulted in 20,640 observations of 9 variables. Those 9 variables are as follows:

- longitude: A measure at how far east/west a house is. A larger value is farther west.
- latitude: A measure of how far north/south a house is. A larger value is farther north.
- housingMedianAge: A measure of the median age of a house in the block. The larger the value, the older the median house is.
- totalRooms: A measure of the total number of rooms within the block.

- totalBedrooms: A measure of the total number of bedrooms within the block
- population: A measure of the total number of people living within the block.
- households: A measure of the total number of households in that block. A household is defined here as a group of people living within one home unit.
- medianIncome: A measure of the median income for households within that block. This is income on a household level and is measured in 10's of thousands.
- medianHouseValue: The median value of a household within the block. This variable is measured in dollars.
- oceanProximity: This is a categorical variable that categorizes houses into five categories: Near Bay, Inland, < 1 Hour Ocean, Island, and Near Ocean.

Additionally, we modified the variable medianHouseValue to be on a more reasonable scale by dividing all values by 100,000. It should be noted that the variables below are measured on the block scale instead of on a household scale.

Results:

We used a simple linear regression model with all of the predictors except latitude and longitude (median house age, log(total rooms), log(total bedrooms), log(population), log(households), log(median income), median house value, and ocean proximity) and included an interaction between log(population) and log(household). While all predictors appeared to be significant, we did not think it made rational sense for there to be a linear relationship between either latitude and longitude and the mean of the median housing prices. Our model gives us the following coefficient estimates and standard errors.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.8	0.16	17	4e-65
ocean_proximityINLAND	-0.69	0.014	-49	0
ocean_proximityISLAND	1.4	0.33	4.4	1.1e-05
ocean_proximityNEAR BAY	0.0024	0.018	0.13	0.89
ocean_proximityNEAR OCEAN	0.095	0.016	5.8	5.5e-09
housing_median_age	0.012	0.00047	26	5.7e-146
log(median_income)	1.6	0.018	88	0
log(total_rooms)	-0.35	0.034	-10	4.2e-24
log(total_bedrooms)	0.64	0.044	14	6.3e-47
log(population)	-1.1	0.03	-35	5e-267
log(households)	0.31	0.047	6.5	9.2e-11
log(population):log(households)	0.04	0.0042	9.6	5.8e-22

Table 2: Fitting linear model: median_house_value_upd ~ ocean_proximity + housing_median_age + log(median_income) + log(total_rooms) + log(total_bedrooms) + log(population) * log(households)

Observations	Residual Std. Error	R^2	Adjusted R^2
20433	0.7286	0.6018	0.6016

Using these values for the beta's and our regression model, an individual who has values for each of our predictor variables could calculate the mean of median house values for that house. While this is something that our model offers, we think that a much more practical use of it is to understand the influence that all the predictors had on housing prices in the particular blocks examined by the census study in California in 1990.

Our model indicated that ocean proximity had a statistically significant effect on the price of homes. Our reference level is houses less than one hour away from the ocean as a reference level. The model found that, after accounting for all other variables, being on an island increased the value of a house from living within one hour of the ocean, houses inland had a lower mean of the median, houses near the bay had a marginally higher mean of median, and we did not find strong evidence that being near the ocean had an effect on housing prices over being within one hour of the ocean.

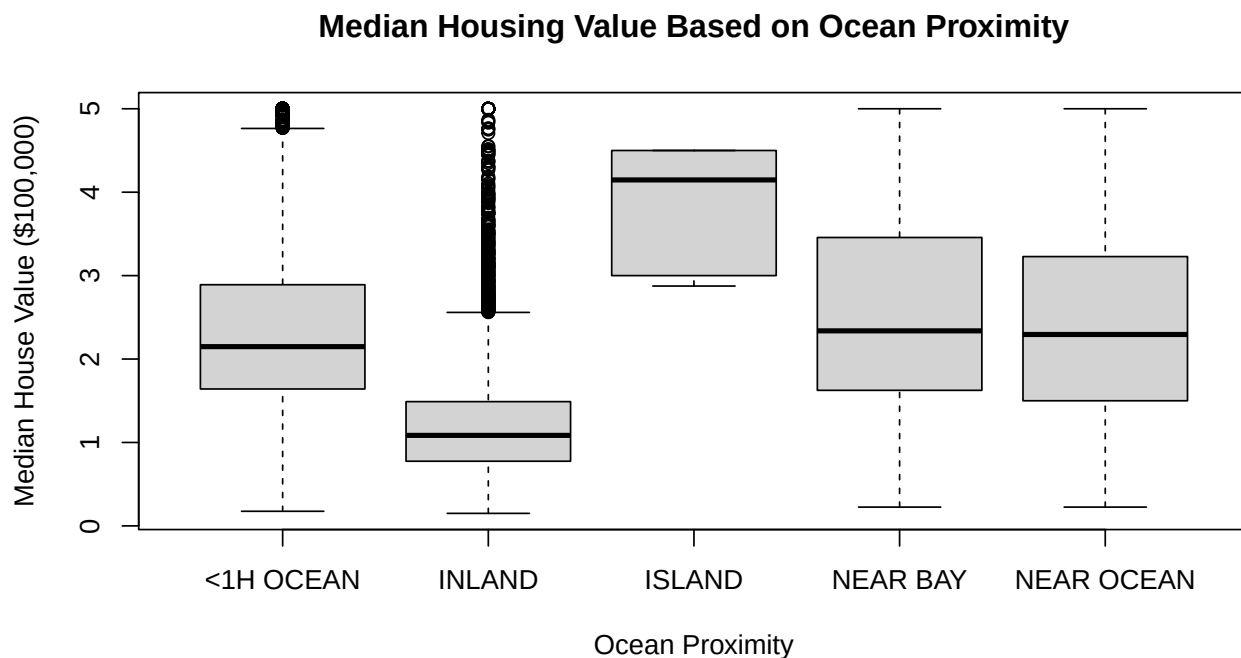
Fixing all other variables, houses inland had a mean median price \$69,000 less than houses within 1 hour of the ocean. We are 95% confident that the mean prices of houses inland were between \$66,256 and \$71,744 less expensive than those within one hour of the ocean.

Houses on an island on the other hand, had a mean median price which was \$140,000 more expensive than those within 1 hour of the ocean, fixing for all other variables. We are 95% confident that, fixing for all other variables, the mean difference in median prices between houses on an island and those within one hour of the ocean was between \$75,320 and \$204,680.

Houses that are near the ocean had a mean median house price \$9,500 more than those within one hour of the ocean after fixing for all other variables. We are 95% confident that, fixing for all other variables, the mean difference between houses near the bay and those within one hour of the ocean is between \$6,364 and \$12,636.

The model found a statistically insignificant p value for the individual t-test of the coefficient for a change in housing prices near the bay. Thus we do not have strong evidence that the prices of houses near the bay were different from those within one hour of the ocean.

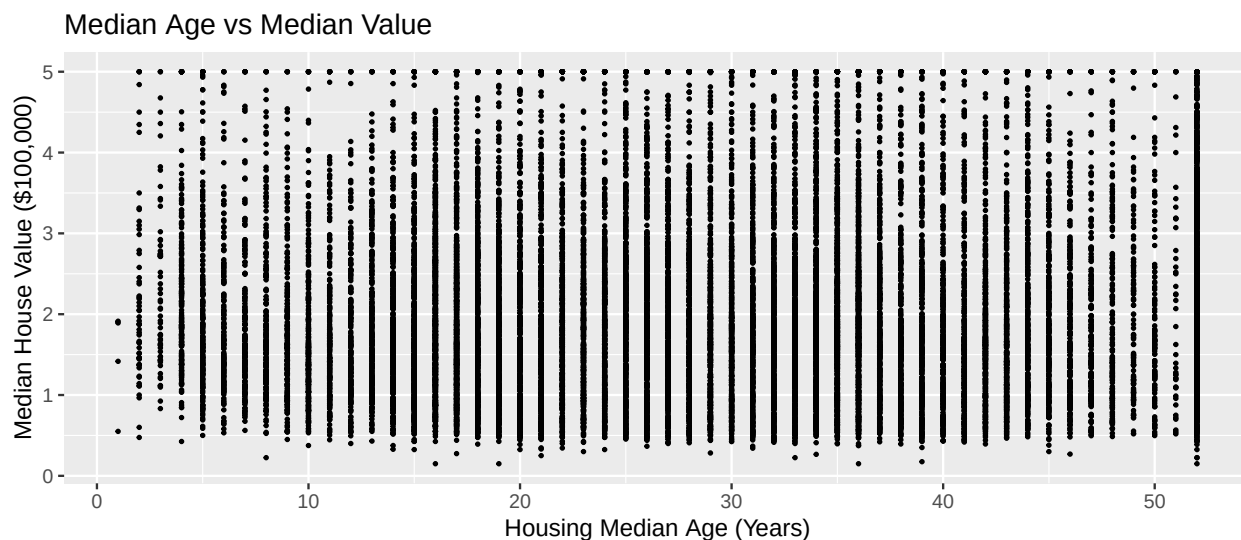
The boxplot below shows the median housing prices based on ocean proximity. While the plot does not account for all other factors, it provides information on which areas have the highest housing prices. We can see, just as our model suggests, that houses on islands are significantly more expensive. Furthermore, we can see that near the bay, near the ocean, and within one hour of the ocean all have relatively similar median house values. Inland houses have a lower median, but what is interesting is the large quantity of outliers on the inland plot. This can likely be attributed to various other factors (like proximity to metropolitan areas and size of houses) that increase the number of outliers inland.



The median age of a house appears to have had a positive effect on housing prices after accounting for all other factors. A one year increase in median housing age was correlated with an \$1,200 increase in value. We

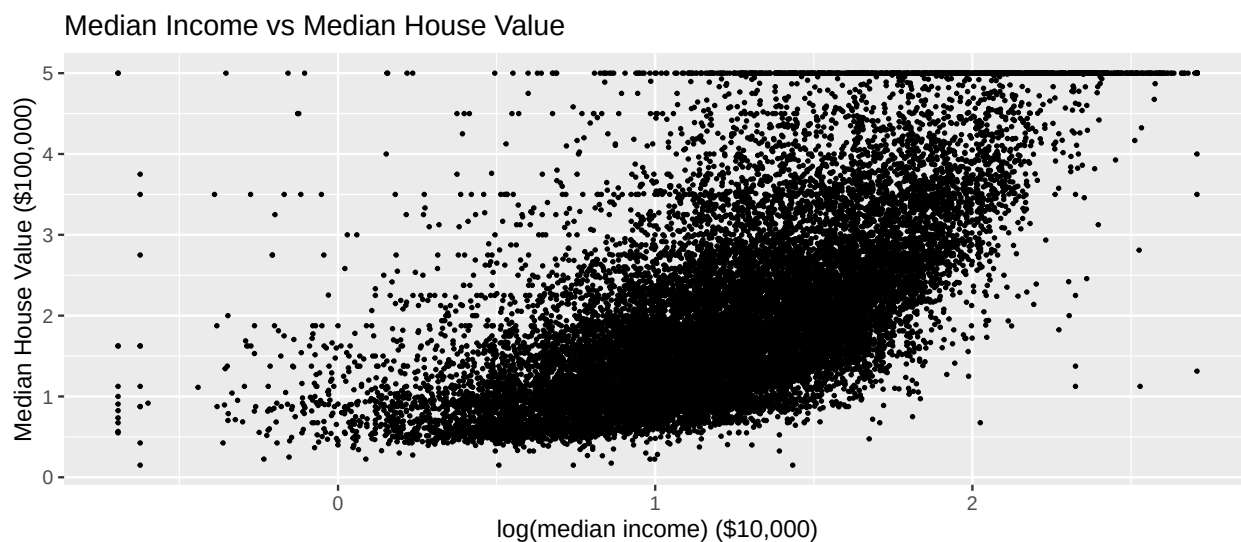
are 95% confident that, after fixing for all other variables, increasing the age of a home by one year resulted in between a \$1,108 and \$1,292 increase in the mean of its median price.

The below plot shows the relationship between the median age of a house and its median value in 1990. We can see that there isn't an easily discernible linear relationship and there is a large scatter throughout. Additionally, we see a large amount of blocks with a median age of 50 years. We suspect, much like with income, this has to do with an upper bound in census reporting.



Median income has a statistically significant effect on the mean of median housing prices. A 10% increase in median income of the block to which a house belongs was correlated with a \$15,250 increase in the mean of the median housing price. We are 95% confident, after fixing for all other variables, that the mean effect fell between a \$15,078 and \$15,421 increase in mean of median house prices.

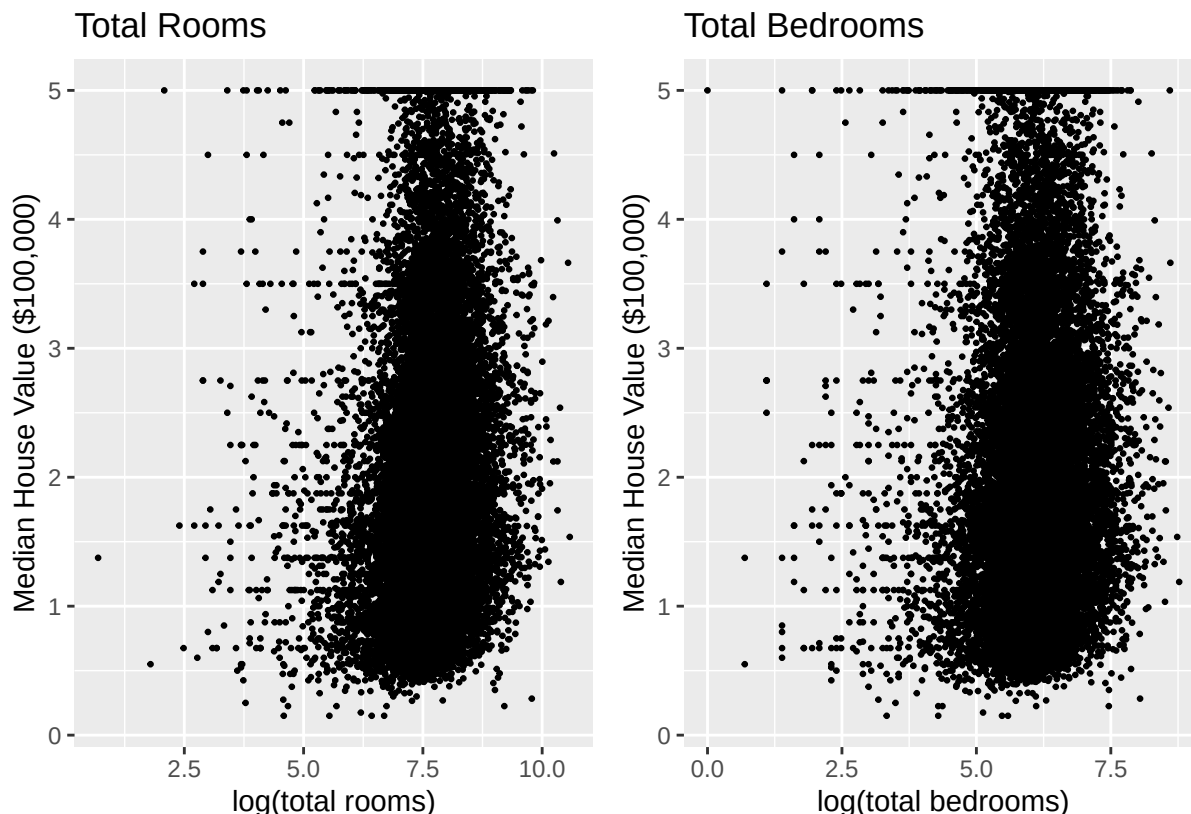
The below plot depicts the relationship between $\log(\text{median income})$ and median house value. While there is a strong scatter, we can see the positive linear correlation between the two variables. This makes intuitive sense because in blocks where there are higher median income, we expect the homeowners to have purchased more expensive houses.



A 10% increase in total rooms in the block was correlated with between a \$3,012 and \$3,660 decrease in mean

of median housing prices after fixing for all other variables (95%). The expected change in mean housing prices was a \$3,336 decrease. But unlike the total number of rooms, which decreased the the average price, a 10% increase in the number of bedrooms was correlated with between a \$5680 and \$6519 increase in housing value after fixing for all other variables (95%).

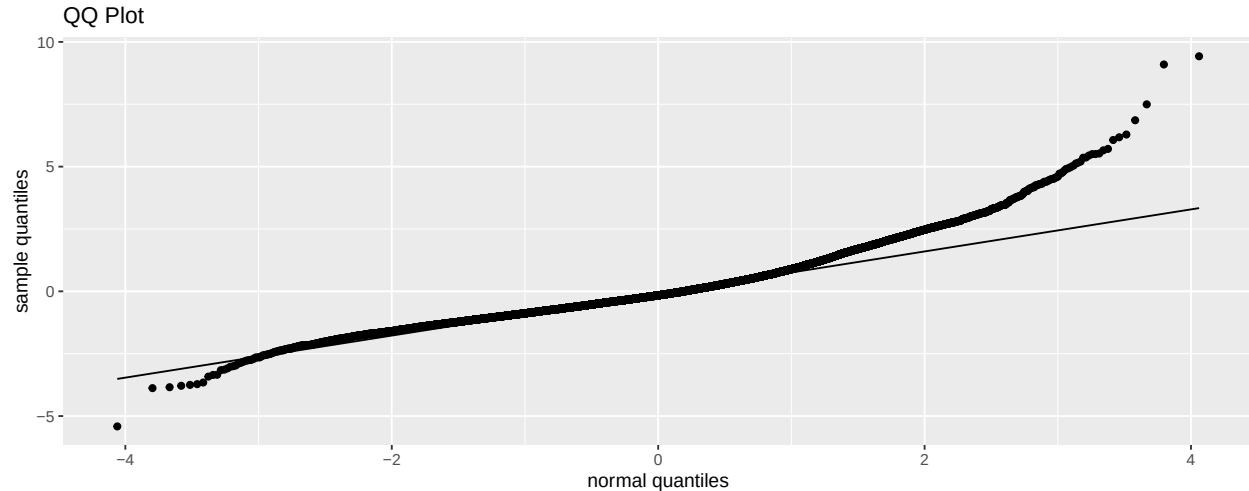
The below plots show the relationship between $\log(\text{rooms})$ and $\log(\text{bedrooms})$ and median house value respectively. As we can see in these plots, there is clustering between 5 and 10 total $\log(\text{total rooms})$ and 4 and 8 $\log(\text{total bedrooms})$ but there appears to be a fairly even scatter throughout and no discernible effect between each predictor individually and the response variable, without accounting for the other predictors.



A 10% increase in just population in the block of the house, fixing all other variables, was correlated with a \$10,480 decrease in the mean of the median price. The 95% confidence interval of that effect goes from a decrease of \$10,198 to \$10,770. A 10% increase in just number of households in the block where the house is located increased the mean of the median house prices by \$2,955. The 95% confidence interval for that effect was \$2,507 to \$3,403. But when there is both a 10% increase in households and population—which is what we would expect—the next effect was a decrease in the mean of the median of \$7,148. The 95% confidence interval for that effect is between a \$6,374 and and \$7,922. This result makes intuitive sense because people value their space and so if more households are built within the same block as a house and the population of the block increases as more people move into the block, we would expect the mean of the median house value to decrease.

We arrived at our model by beginning with the saturated model and testing for reasonable pairwise correlations. Once we removed all statistically insignificant correlations and determined which coefficients to drop through using AIC and our own logic (such as removing longitude and latitude because there did not logically appear to be a linear correlation between them and mean of median housing price), we made the necessary transformations to bring all predictor variables to a linear scale. A number of the variables had a fanning out towards the lower end and we applied a log transformation to fix the heteroskedacity to align the predictor variables with our assumptions for a simple linear regression model. Our final QQ plot can be seen below the two plots. There appears to be some heavy tailed-ness on the right end—which is to be expected with

housing prices. We are not concerned by that because of the size of our dataset.



Discussion:

We found that being on an island, near an ocean, being older, having higher median income, and have more total bedrooms all increased the mean of the median of housing prices. Being inland, having more total rooms in a block, and having a higher population, all reduced the mean of the median price of a house in a block. It was clear from our model that for many of these predictors, they did not have a clear linear relationship with the response variable without accounting for the other variables, but when all compiled, provide us with a regression model that can help predict those housing prices in California in 1990. This makes intuitive sense because housing valuation is deeply nuanced. Houses in more sought after areas are smaller and have less people and yet are more expensive, but apartments which have lots of people in a busy downtown space are also quite expensive. Just looking at the number of people can't give us an appropriate understanding of its influence on housing prices. This is why we saw a large scatter in almost every variable. The two variables that were easier to see were the proximity to the ocean and median income. Both of these have easier effects to examine separate from the broader model because land closer to the ocean is more limited and thus more sought after and people with a higher median income, and thus more disposable income, will likely use that disposable income to purchase a more expensive house.

There were a number of limitations in our analysis that can be further examined in future projects. The first is that we limited ourselves to pairwise interactions and thus were not exhaustive in our analysis of interactions. It is possible that there are interaction effects that exist outside the scope of the interactions we examined. Furthermore this data was self-reported in a census from 1990. This provides a couple of limitations. First, this data is not applicable to analysis of housing prices in California today. It is strictly limited to 1990 and the specific blocks examined by the census. The second is that since this data is self reported it is possible that all the data may not be accurate. More advanced statistical methods (such as conducting a median regression instead of a mean) might allow us to better examine the effects of these predictors on the data.

Ultimately, this examination left us with a number of questions. We remain curious about why there isn't a statistically significant interaction between total number of bedrooms and total number of rooms. We remain curious about the inclusion of variables like latitude and longitude which require spacial data. We remain curious about conducting time-series analysis on this data to see how the effects of these predictors have changed over time. Our R^2 value indicates that about 60% of the variability we see is explained by our model, so there is clearly room to include additional variables like proximity to major cities or additional interactions. There is a lot to explore when it comes to housing prices, and this model just begins to explore the depth of examination that one can do with more data and a greater knowledge of statistical tools.

References:

Data Source - <https://www.kaggle.com/camnugent/california-housing-prices>

Average Housing Price 1990 - <https://tinyurl.com/yh2v8ssk>

R code Appendix